

The four subspaces, by matrix type

Three matrices with the same four subspaces computed for each. The point is the pattern across them — how much of each space collapses as the rank falls.

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CASES

TYPE	CONDITION	COL(A)	ROW(A)	NULL(A)	NULL(A ^T)	
NOTHING COLLAPSES						
	Invertible square	$r = n = m$	all of R	all of R	$\{0\}$	$\{0\}$
ALMOST EVERYTHING COLLAPSES						
	Rank one	$A = \mathbf{u} \mathbf{v}^T, r = 1$	the line through $\mathbf{u}$	the line through $\mathbf{v}$	hyperplane, dim = $n - 1$	hyperplane, dim = $m - 1$
COLLAPSE SPLIT EVENLY						
	Symmetric projection	$A = A^T, A^2 = A$	the target subspace S	the same as Col(A)	S	the same as Null(A)

Rank is the only variable. At $r = n$ nothing is lost and both null spaces are trivial; at $r = 1$ almost everything is lost and the null spaces are hyperplanes. Every row obeys $r + \text{nullity} = n$ on the domain side and $r + \text{left nullity} = m$ on the codomain side — the **dimension accounting** with three sets of numbers substituted in.